

RubricGuard

AI-Powered Grading Consistency & Fairness Engine

Hackathon Pitch 2026

Meet Prof. Sharma

He teaches BUS302 — Strategic Management. 15 students across 3 sections, graded by 3 professors. An argumentative essay on whether universities should ban laptops in lectures.

Prof. Sharma

Section A · 5 students

← You are here

Prof. Somani

Section B · 5 students

Prof. Raymahesh

Section C · 5 students

THE PROBLEM

Grading is **broken**.

Every semester, professors grade thousands of essays with no systematic way to ensure consistency, fairness, or evidence-based justification.

8–12%

of assignments receive
regrade requests

10–15%

avg score variance on
equivalent work

70–80%

rubric adherence rate vs.
≥95% target

Subjective scoring with no accountability

Graders assign scores based on gut feeling. Justifications are vague, rarely tied to specific evidence.

- ⚠ Justifications like 'good argument' with no specific evidence cited
- ⚠ Same quality work receives 15/25 from one grader and 22/25 from another
- ⚠ No mechanism to verify if scores actually match rubric descriptors



"Well-written essay.
Good arguments."

Score: 22/25 – but why?

STU001 **18/25**

Sharma's Section A

STU006 **24/25**

Somani's Section B

6-point gap

for essays citing the same research

Invisible fairness violations

When students across sections write essays of comparable quality, there's no system to detect when they receive wildly different grades.

- △ Cross-section consistency is never checked
- △ Grader bias goes completely undetected
- △ Students lose trust in the fairness of grading

It's 2 AM. Prof. Sharma is on his 12th essay.

He wonders: "Am I scoring STU005 the same way I scored STU001 three hours ago?" He checks Canvas — no help. Gradescope — no consistency metrics. Turnitin — only plagiarism.



Canvas

No consistency checking



Gradescope

No fairness detection



Turnitin

No score validation

He needs a copilot, not another grading tool.

Existing tools grade. **None validate.**

Canvas SpeedGrader

- ✓ Inline comments
- ✓ Rubric templates
- ✓ Speed tools

✗ No consistency
checking

Gradescope

- ✓ AI grouping
- ✓ Batch grading
- ✓ Rubric reuse

✗ No fairness detection

Turnitin

- ✓ Plagiarism detection
- ✓ Grammar check
- ✓ Similarity

✗ No score validation

RubricGuard

- ✓ AI validation ✓
- ✓ Fairness detection ✓
- ✓ Live analytics ✓

ENTER RUBRICGUARD

What if every score was defensible?

An AI copilot that validates every justification, detects fairness violations across students, and gives professors real-time visibility into grading quality — without replacing human judgment.

Evidence-Based Scoring

Attach specific text excerpts as evidence for each criterion

AI Validation

Every justification validated against evidence in real time

Fairness Detection

Pairwise comparison flags similar answers with different scores

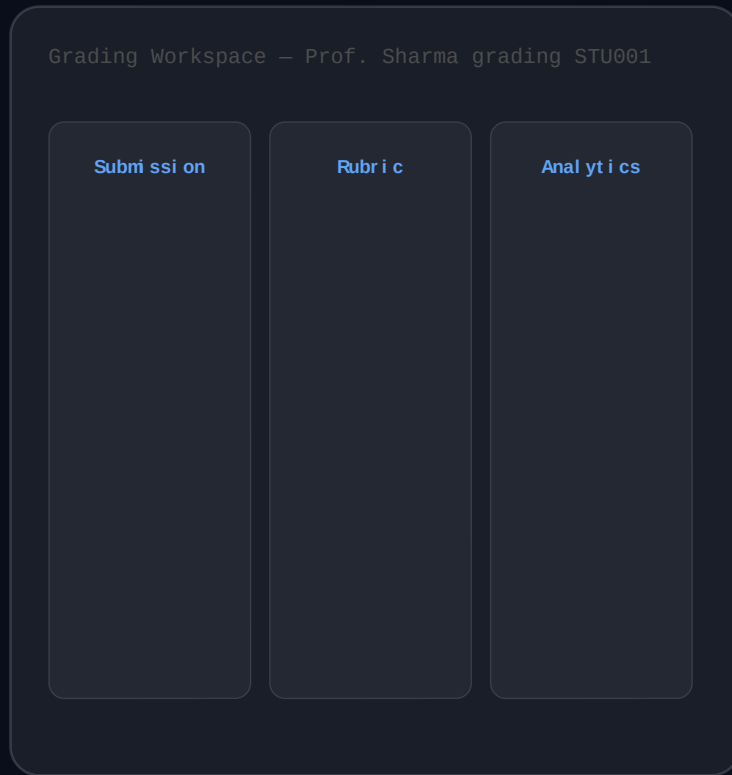
Segment Grades

Cross-grader analytics with per-professor validity rates

3-Column Grading Interface

Prof. Sharma opens RubricGuard and begins grading. The workspace shows submission, rubric, and live analytics side-by-side.

- Left: Student submission with text selection & highlighting
- Center: Rubric criteria with scores, justifications, and AI suggestions
- Right: Live analytics updating with each grading action
- Dual-color highlighting: blue (user) + yellow (AI key quotes)



Every score evaluated against AI

For each rubric criterion, the AI independently reads the submission and generates a suggested score with detailed reasoning.

- AI reads the full submission and scores each criterion independently
- Suggested score shown as collapsible bar (e.g., 'AI SUGGESTION 24/25')
- Expand to see AI's detailed reasoning and supporting evidence quotes
- 'Accept' adopts AI score, or 'Keep My Score' to stay with yours

Argument Clarity

Score: 22 / 25

Selected Evidence: "Universities should implement a carefully structured laptop policy..."

✦ AI SUGGESTION 24/25

The 3-mark deduction seems harsh — the student's thesis progressively narrows from general to specific policy.

✓ **Accept (24/25)** ✗ Keep My Score

Validation Pipeline

1 Grader highlights evidence & writes justification

2 AI cross-checks evidence × justification × rubric

3 Verdict: Fully / Partially / Not Supported

AI validates the grader's work — it never assigns the grade itself

AI VALIDATION

"Is my justification backed by evidence?"

Prof. Sharma scores STU002's Argument Clarity at 12/25. He clicks Validate — the AI checks if his evidence supports that deduction.

□ Fully Supported — justification matches the evidence

△ Partially Supported — some gaps in reasoning

□ Not Supported — evidence doesn't back the score

AI extracts Key Quotes and suggests score refinements

AI feedback that teaches, not just judges

Clicking a validation status badge opens the AI's full reasoning — evidence found, gaps identified, and specific suggestions.

- Referenced submission evidence shown with exact quotes
- Detailed AI reasoning explains why support level was assigned
- Suggested refinements help graders write stronger justifications
- Key Quotes highlight the most criterion-relevant passages

⚠ **Partially Supported** Evidence Use · STU002

YOUR EVIDENCE

"Mueller & Oppenheimer (2014) found that longhand notes improve conceptual learning..."

AI REASONING

Your justification mentions "limited sources" but the student cites 4 papers. The issue is depth, not quantity.

SUGGESTED REFINEMENT

"While citing Mueller & Oppenheimer, the discussion doesn't address methodological limitations..."

KEY QUOTES FOUND

"Sana, Weston, and Cepeda (2013) found that laptop multitaskers perform 11% worse..."

Explanation Validity Rate

A single metric that captures grading quality — how well justifications hold up against AI validation, with a penalty for fairness violations.

1 **Count validations**
$$\text{fully_supported} / \text{total_validated} \times 100$$

2 **Fairness penalty**
$$\text{fairness_flags} \times 1\% \text{ (capped at 20\%)}$$

3 **Final rate**
$$\text{raw_validity} - \text{fairness_penalty}$$

Prof. Sharma (5 students × 4 criteria)

Fully Supported 15

Partially Supported 3

Not Supported 2

Raw: $15/20 = 75\%$ Penalty: $1 \text{ flag} \times 1\% = 1\%$

Final Validity Rate **74%**

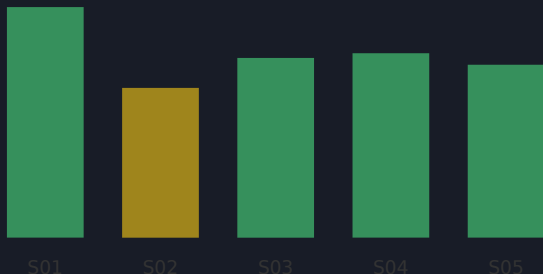
Live Analytics — Right Panel

Session Progress 5/5

Validation: Fully 4 | Partial 0 | Not 0

Validity Rate **75%**

$85\% - 10\% = 75\%$



LIVE DASHBOARD

Real-time analytics that update as you grade

The right panel is a live dashboard — every score, validation, and fairness check updates instantly.

Session Progress

Tracks students fully graded

Validation Quality

Fully / Partially / Not Supported breakdown

Explanation Validity Rate

Raw validity minus fairness penalty

Validity Rate Trend

Line chart plotting cumulative validity per student

Grading Fairness Alert

The AI independently assesses each answer's quality. When two students receive similar AI assessments but different grader scores, a fairness alert fires.

- AI scores each criterion independently (aiSuggestedScore)
- Compares all student pairs: $|aiDiff| \leq 2$ = 'similar quality'
- If $|scoreDiff| > 1.5$ on similar quality → flag inconsistency
- Each flag penalizes Validity Rate by 2% (capped at 20%)

```
if ( |AI_A - AI_B| ≤ 2 && |Score_A - Score_B| > 1.5 ) → FLAG
```

Fairness Alert — Argument Clarity

Argument Clarity 6 pt gap

STU001 AI: 22 18/25

STU004 AI: 23 24/25

⚠ AI says similar quality (diff: 1), but scores differ by 6 pts

Session Analytics — Prof. Sharma

[Grading Session Report]
Score Distribution · Timeline · Heatmap

SESSION REPORT

Grading Session Analytics

After grading 5 students, Prof. Sharma reviews his session report: quartile analysis, score timeline, criterion stability, heatmap, and fairness audit.

- Score distribution with mean, median, Q1/Q3, and IQR
- Per-criterion standard deviation shows scoring stability
- Color-coded heatmap across all 5 students × 4 criteria
- Consolidated fairness alerts with pairwise comparisons

DEEP DIVE

Heatmap, Timeline & Stability

Score Timeline

Area chart showing total scores across sequence

Criterion Stability

Bar chart of std dev per criterion

Criterion Heatmap

Student × Criterion matrix, color-coded

Grading Fairness Alerts

Aggregated pairwise comparisons

Score Heatmap — All Students × Criteria

	Arg.	Evid.	Anal.	Writ.
STU001	22	19	21	23
STU002	22	21	8	15
STU003	24	24	25	24
STU004	21	23	20	22
STU005	16	22	19	21

CROSS-SECTION

Segment Grades: 15 students, 3 graders

The Segment Grades page shows a cross-section comparison — all three professors side by side with their validity rates, fairness alerts, and validation breakdowns.

Pr of . Shar ma

Section A · 5 students

82 Avg 74% Valid 1 Alerts

Your section

Pr of . Somani

Section B · 5 students

79 Avg 88% Valid 0 Alerts

Pr of . Raymahesh

Section C · 5 students

85 Avg 67% Valid 2 Alerts

Segment View — Prof. Sharma Detail

[Score Table + Validity Trend]

SEGMENT INSIGHTS

Per-grader deep dive with actionable insights

- Validity Rate Trend per grader — spot who's drifting
- Score table with color-coded cells per criterion
- Status dots: green = fully, yellow = partial, red = not supported
- Fairness alert count & validity rate badges
- Cross-section comparison across all 3 professors

Storyboard

1 Launch

Open RubricGuard

2 Select Assignment

Choose BUS302 CA1

3 Enter Workspace

3-column interface

4 Review & Highlight

Select evidence text

5 Score & Justify

Enter score + reasoning

6 Validate

AI checks justification

7 Consistency Alert

Fairness flag if drift

8 Adjust / Override

Revise or keep score

9 Finalize & Review

Session analytics

RubricGuard supports human judgment —it does not replace it.

FOR PROFESSORS

Cross-Section Oversight

Prof. Sharma can see how Somani and Raymahesh are grading. One metric — Explanation Validity Rate — tells the whole story.

- Per-grader validity rates at a glance
- Cross-section fairness highlights scoring drift
- Criterion-level breakdown shows WHERE each grader is lenient/strict
- Validation breakdown: fully / partially / not supported counts

Explanation Validity Rate by Grader

Prof. Somani (B) 88%



Prof. Sharma (A) 74%



Prof. Raymahesh (C) 67%



Flagged Issues

⚠ Raymahesh: 2 alerts — Evidence Use inflated

⚠ Sharma: 1 alert — Argument Clarity inconsistency

Improvement Over Time

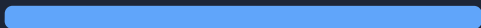
Assignment 1

54%



Assignment 3

71%



Assignment 6

88%



GROWTH LOOP

Every grader gets better

RubricGuard isn't just monitoring — it's a training system. Each validation teaches what a well-justified score looks like.

- Real-time AI feedback on every justification
- Suggested refinements teach evidence-based reasoning
- Validity rate trends show personal growth over time
- Fairness alerts highlight specific blind spots

Dual Highlighting as a Teaching Moment

When your blue highlights don't overlap with AI's yellow key quotes, that gap becomes a learning signal.

- Blue = what YOU thought was important
- Yellow = what the AI identified as most relevant
- Overlap = strong alignment with rubric intent
- Gap = a blind spot worth examining

Evidence Use – STU003

YOUR HIGHLIGHT

"Mue l l e r & O p p e n h e i m e r s h o w e d l o n g h a n d n o t e s i m p r o v e c o n c e p t u a l u n d e r s t a n d i n g . . . "

AI KEY QUOTE

"H o w e v e r , t h e s t u d y ' s l a b c o n d i t i o n s m a y n o t g e n e r a l i z e , a s n o t e d b y M b r e h e a d e t a l . (2 0 1 9) . . . "

Learning insight: You highlighted supporting evidence but missed the counter-argument. Recognizing nuance is key.

First-Time Grader Calibration

First Essay

42%

No evidence cited, vague justification

Third Essay

68%

Citing passages, but missing counter-arguments

Fifth Essay

91%

Evidence-rich, rubric-aligned, defensible

CALIBRATION

New TAs calibrate 3× faster

First-time graders go from vague 'good essay' feedback to evidence-rich, rubric-aligned scoring within 5 essays.

- AI provides immediate, specific feedback on each justification
- Graders learn what 'strong evidence' looks like through the AI's lens
- Validity rate trajectory shows personal improvement over time
- Traditional: 2–3 weeks to calibrate. RubricGuard: 5 essays

BIGGER PICTURE

From individual grading to institutional learning

RubricGuard creates a feedback loop that improves grading quality at every level — individual graders, course sections, and entire departments.

Individual

Every grader calibrates faster with AI feedback

Section

Cross-student fairness detection within each section

Department

Cross-grader analytics ensure institution-wide standards

IMPACT

Measurable impact

Regrade Reduction

↓ **60%**

Fewer appeals when
every score is evidence-
backed

Grader Calibration

3× faster

New TAs reach
competency in 5 essays
vs. 2-3 weeks

Score Variance

↓ **40%**

Cross-section scoring
variance reduced
significantly

Time Saved

2
hrs/week

Less time on regrade
requests and grade
disputes

TECH STACK

Built for scale, designed for trust

React + TypeScript

Frontend framework

Supabase

Backend & auth

Gemini AI

Validation engine

Recharts

Data visualization

Tailwind CSS

Design system

Vercel

Cloud deployment

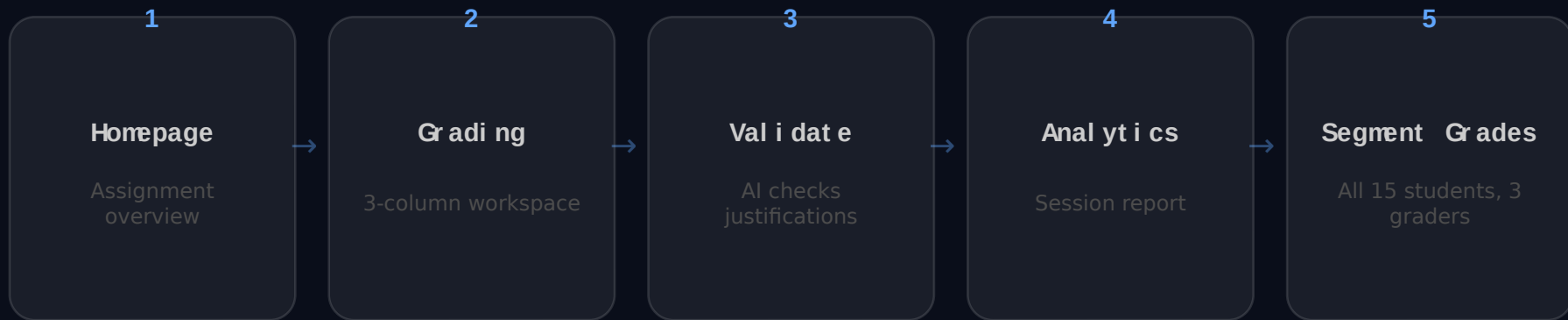
29 slides

6+ live screens

100% cloud-native

LIVE DEMO

The Flow You'll See



Homepage → Grade 5 students → View Analytics → See Full Segment Grades

Grade with confidence.

Every score justified. Every justification validated.
Every student graded fairly.

↗ Live Demo Available ↘

Thank you — Questions?